Paper ID: 16280 **Roll No.............................**

END-SEMESTER EXAMINATION, 2024-25
(TERM : 2401)
CSE 249 : DATA BASE MANAGEMENT SYSTEM

Time : 03 Hrs. **Max. Marks : 100**

Note: 1. **All** questions are compulsory.
2. Assume missing data suitably, if any.

CO1: Explain the basics concepts of data base.
CO2: Demonstrate the knowledge of databases to E-R modelling.
CO3: Ability to design entity relationship and convert entity relationship diagrams into RDBMS and formulate SQL queries on the respective data
CO4: Apply normalization techniques to reduce redundancy from the database.
CO5: To appraise the basic issues of Transaction processing, Serializability& concurrency control
CO6: Design & develop database for real life problems

SECTION-A

All Questions are Compulsory: **(10×4=40 Marks)**

S.No.	Question	COs	Marks	BTL
1.	What are attribute and Entity? Give examples.	CO1	4	K1
2.	What are Advantages and Disadvantages of DBMS?	CO1	4	K2
3.	Explain how to create table with syntax in SQL.	CO2	4	K1
4.	Explain following terms i) DBMS ii) Primary Key iii) Candidate Key	CO2	4	K2
5.	Explain Boyce Codd Normal Form.	CO3	4	K2
6.	Differentiate between Trivial and Non-Trivial Functional Dependency.	CO3	4	K3
7.	What are Schedules?	CO4	4	K2
8.	Explain ACID properties of transaction processing system.	CO4	4	K3
9.	Explain two phase locking protocol with suitable example.	CO5	4	K2
10.	Differentiate between strict and basic timestamp ordering protocol	CO5	4	K3

SECTION-B

All Questions are Compulsory: **(3×6=18 Marks)**

S.No.	Question	COs	Marks	BTL
11.	(a) Define all the aggregate functions in SQL with example.	CO3	6	K4
	---- OR ----			
	(b) List the Armstrong axioms for functional dependencies.			
12.	(a) Describe major problem associated with concurrent processing with example.	CO4	6	K4
	---- OR ----			
	(b) Check whether the given schedule S is conflict serializable or not- S : R1(A) , R2(A) , R1(B) , R2(B) , R3(B) , W1(A) , W2(B)			

13. (a) What are two phase locking protocol? List the salient feature of strict two phase locking protocol. CO5 6 K4

---- OR ----

(b) Explain validation based protocol in concurrency control with example.

SECTION-C

All Questions are Compulsory: **(3×10=30 Marks)**

14. (a) Given R(,A ,B ,C ,D E) with the set of FDs, F{AB → CD,A → ,E C → D}. Is the decomposition of R into R1 (A ,B C) ,R 2 (B ,C D) and R3 (C ,E,D) lossless? CO3 10 K4/K5

---- OR ----

(b) What is Normalization? Write the advantages and disadvantages of normalization.

15. (a) Analyze serial and non serial schedule using suitable example. CO4 10 K4/K5

---- OR ----

(b) Write the procedure to check a given schedule is conflict or not.

16. (a) What is log? Compare differed database modification and immediate database modification technique of log base recovery with example. CO5 10 K4/K5

---- OR ----

(b) Discuss the problems of deadlock and starvation, and the different approaches to dealing with these problems.

SECTION-D

Attempt the following Question: **(1×12=12 Marks)**

17. (a) Draw ER diagram for banking sector and convert the given ER diagram to relational table CO6 12 K5/K6

---- OR ----

(b) Write query for (i) Create table (ii) Insert record in existing table (iii) update record in a table (iv) Create primary key (v) Second maximum salary from a given table.

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